

Geophysical Inversion

Problem Set Week 1

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Problem 1.1

Suppose that you determine the masses of 100 objects by weighing the first, then weighing the first and second together, and then weighing the rest in triplets: the first, second, and third; the second, third, and fourth; and so forth.

A

Identify the data and model parameters in this problem. How many of each are there?

Solution

The data is the measurements of the triples. The model parameters are estimates of the real weight of the 100 objects. There are 100 of each.

B

Write down the matrix $\mathbf{G}$ in the equation $\mathbf{d} = \mathbf{G}\mathbf{m}$ that relates the data to the model parameters.

Solution

$$\mathbf{G} = \begin{bmatrix} 1 & 0 & 0 & 0 & \cdots & \cdots & \cdots & \cdots & 0 \\ 1 & 1 & 0 & 0 & & & & & \vdots \\ 1 & 1 & 1 & 0 & & & & & \vdots \\ 0 & 1 & 1 & 1 & & & & & \vdots \\ \vdots & & & & \ddots & & & & \vdots \\ \vdots & & & & & & 1 & 1 & 1 & 0 \\ 0 & \cdots & \cdots & \cdots & \cdots & 0 & 1 & 1 & 1 \end{bmatrix}$$

C

How sparse is $\mathbf{G}$? What percent of it is zero

Solution

Approximately 97% is zero. For a more precise estimate that takes the first two lines into account; we have:

$$1 - \frac{\frac{1}{100} + \frac{2}{100} + 98 \times \frac{3}{100}}{100} = 1 - \frac{297}{100^2} = \frac{9703}{10000}$$

This means that 97.03% of the matrix is zero.

Problem 1.2

Suppose that you determine the height of 50 objects by measuring the first, and then stacking the second on top of the first and measuring their combined height, stacking the third on top of the first two and measuring their combined height, and so forth.

A

Identify the data and model parameters in this problem. How many of each are there?

Solution

The data are the height measurements of the different stacks. The model parameters are estimates of the heights of individual objects. There are 50 of each.

B

Write down the matrix $\mathbf{G}$ in the equation $\mathbf{d} = \mathbf{G}\mathbf{m}$ that relates the data to the model parameters.

Solution

$$\mathbf{G} = \begin{bmatrix} 1 & 0 & 0 & 0 & \cdots & 0 \\ 1 & 1 & 0 & 0 & & \vdots \\ 1 & 1 & 1 & 0 & & \vdots \\ 1 & 1 & 1 & 1 & & \vdots \\ \vdots & & & & \ddots & \vdots \\ 1 & 1 & 1 & 1 & \cdots & 1 \end{bmatrix}$$

C

How sparse is $\mathbf{G}$? What percent of it is zero

Solution

We can immediately see that slightly less than half of the matrix will be zero. For a more precise estimate we can see that line n contains the number 1 n times. To get the total number of 1s we can use the formula $\frac{n(n+1)}{2}$ to add up the numbers from 1 to 50 and then divide by the total number of elements: 50^2 .

$$\frac{\frac{50 \times 51}{2}}{50^2} = \frac{1275}{2500} = 0.51$$

Therefore $100 - 51 = 49\%$ of the matrix is zero.

Problem 1.3

Write a program to compute $\mathbf{G}$ in the case of the cubic equation, $T = a + bz + cz^2 + dz^3$. Assume that 11 z s are equally spaced from 0 to 10.

Solution

```
# Compute the G matrix for a cubic polynomial model given vector z.
# z is a vector
# z is represented either as a 1D array or a 2D array column vector (z.shape[1] == 1)
def ComputeG(z):
    n = z.shape[0]

    #In case z is a 1D array
    if len(z.shape) == 1:
        z = z.reshape(n,1)

    G = np.concatenate((np.ones((n,1)), z, z**2, z**3), axis = 1)
    return G

z = np.arange(11)
print(ComputeG(z))
```

Output:

```
[[ 1.  0.  0.  0.]
 [ 1.  1.  1.  1.]
 [ 1.  2.  4.  8.]
 [ 1.  3.  9. 27.]
 [ 1.  4. 16. 64.]
 [ 1.  5. 25. 125.]
 [ 1.  6. 36. 216.]
 [ 1.  7. 49. 343.]
 [ 1.  8. 64. 512.]
 [ 1.  9. 81. 729.]
 [ 1. 10. 100. 1000.]]
```

Problem 1.5

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